

Challenges related to Ducts, Chimneys and Tail-end Alignment

Online health monitoring of Suction Ducts

Inspection of suction ducts carrying exhaust fumes, dust & gases is carried out periodically to assess their condition. Currently physical checks are being carried out to assess the condition of these ducts and ultrasound probes are also used to measure wall thickness. However, inspecting these ducts poses challenges, such as calling for a complete shutdown, and inspection at height which poses safety hazard. Some ducts are also part of the processes which remain under a negative suction pressure ranging from 110 to 120 mbar and operate at a temperature range of 120 to 180 degrees Celsius.

We are seeking an online inspection system, preferably portable or capable of covering the entire length of duct. Ideally, this system should conduct measurements (least count of measurement should be $<0.1\text{mm}$) during plant operations, eliminating the need for shutdowns.